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AES standard for metadata –

Processing and handling history

of audio objects

Abstract

This standard provides a vocabulary to be used in describing structural and administrative metadata about the processing and handling history of audio objects during audio preservation activities. Some implementations also refer to this metadata as technical metadata. The characteristics of the processing and handling of audio objects captured by this standard may be of use to audio communities beyond the audio preservation community.

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NOTE This filename will be confirmed on publication on this standard, and may be updated in future versions of this standard. 26

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Foreword

This foreword is not part of the AESxx-xxxx AES standard for metadata – Processing and handling history of audio objects.

Work on this project began in earnest in 2002 following a series of meetings that the chair attended at the Library of Congress in which the subject of technical metadata requirements for audio were discussed. Since that time, Harvard University developed a working proof of concept of the metadata schema herein described, which was shared with Indiana University as a part of the National Endowment for the Humanities funded Sound Directions research project. During that project, IU developed their own prototype tools that implemented that earlier draft of the

schema. This version retains all of the earlier functional requirements of the earlier draft while offering a somewhat simplified implementation.

The members of the writing group that developed this document in draft are:

David Ackerman

Chair, working group SC-03-06

Note on normative language

In AES standards documents, sentences containing the word “shall” are requirements for compliance with the document. Sentences containing the verb “should” are strong suggestions (recommendations). Sentences giving permission use the verb “may”. Sentences expressing a possibility use the verb “can”.

DRAFT

AES standard for metadata –

Processing and handling history

of audio objects

Introduction

This standard sets out the vocabulary to be used in describing the processing and handling history of audio objects in audio preservation work. This metadata, sometimes referred to as ‘process history’, ‘digital provenance’ (or ‘digi-prov’), describes the details of how audio objects are treated and/or migrated. Essentially it provides the ‘who’, ‘where’, ‘what’, ‘when’ and ‘how’ information that clarifies the provenance of the audio object with regard to the work that has been done to restore and/or preserve the object. This vocabulary is expressed in the form of an Extensible Markup Language (XML) schema. By doing so it provides a structured human readable instance document that can be easily parsed and manipulated using any of a number of freely available programming libraries and tools.

0 Preamble

0.1.1 Decimal points

According to IEC directives, the comma is used in all text to indicate the decimal point. However, in the specified coding, including the examples shown, the full stop is used as in IEC programming language standards.

0.1.2 Data representation

In this standard, all coding and data representations are printed in an equally-spaced font. Inherited elements and attributes are printed in an italicized equally-spaced font.

0.1.3 Case sensitivity

All element, attribute, data type and constant names shall be case sensitive.

0.1.4 Logical conventions and data types

In this standard, all elements, attributes and custom data types are defined in the lower-camel-case coding convention. For example, lowerCamelCase.

This standard is implemented as an Extensible markup language (XML) schema according to the W3C XML schema recommendation. The full listing in the schema format is provided in this documents appendices.

This standard relies on a number of data types that are defined by the W3C. See Clause 2 Normative references. W3C defined datatypes appear in the text prefixed with the 'xsd:' namespace qualifier. The following W3C data types are used:

xsd:dateTime

xsd:ID

xsd:IDREF

xsd:IDREFS

xsd:positiveInteger

xsd:string

This standard also relies on data types defined in AES57-2011 AES standard for audio metadata – Audio object structures for preservation and restoration. AES57 datatypes appear in the text prefixed with the 'aes57:' namespace qualifier. The following AES57 datatypes are used:

aes57:audioObjectType

aes57:timeRangeType

This standard also relies on data types defined in AES60-2011 AES standard for audio metadata – AES standard for audio metadata – Core audio metadata. AES60 datatypes appear in the text prefixed with the 'aes60:' namespace qualifier. The following AES60 datatypes are used:

aes60:editUnitNumberType

aes60:contactDetailsType

Finally, this standard defines a number of custom data types according to the rules of the W3C XML schema recommendation. These types are defined throughout the text as they are encountered in the element and attribute definition listings.

1 Scope

This standard concerns the technical documentation of the processing and handling of audio files and physical audio formats carried out during audio preservation and restoration work. This documentation, here referred to as metadata, provides a description of where the work occurred, who did the work, when the work was completed, and how the work was done, including a full description of the audio signal chain as it was configured during the processing.

2 Normative references

The following standards contain provisions which, through reference in this text, constitute provisions of this document. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this document are

encouraged to investigate the possibility of applying the most recent editions of the indicated standards.

AES57-2011, AES standard for audio metadata – Audio object structures for preservation and restoration; Audio Engineering Society, New York, NY., US.

AES60-2011, AES standard for audio metadata – Core audio metadata; Audio Engineering Society, New York, NY., US.

W3C XML Schema Part 1: Structures Second Edition, W3C Recommendation 28 October 2004.

W3C XML Schema Part 2: Datatypes Second Edition, W3C Recommendation 28 October 2004.

W3C XML Linking Language (XLink) Version 1.1: W3C Working Draft 07 July 2005.

I think we need at least an Informative Reference to the BWF (aka AES 31-2) here, partly because we are talking about the “handling of audio files” and partly because Figure 1 makes direct reference to it. Further, we ought to cite EBU Tech 3285 Supplement 5 (HYPERLINK "<http://tech.ebu.ch/publications/tech3285S5>"<http://tech.ebu.ch/publications/tech3285S5>), for that is a most useful way to embed the XML in the audio file itself, particularly when sending a file somewhere.

3 Definitions and abbreviations

3.1 Audio object

A general term used to describe physical audio carriers, and digital audio in the form of files and streams.

For the purposes of this document, the term physical audio carriers shall be understood to mean only those carriers to which an audio data essence is intrinsically bound to the physical object. For example, an audio DAT tape is a physical digital audio object. Wave files on the other hand are file based digital audio objects. They can be moved, copied, and renamed without the need to play the audio essence. In the latter case, the file exists on a storage device, but it is not intrinsically bound to that device. Therefore there is no need to describe the physical storage given its transient nature.

For additional details see AES57-2011, AES standard for audio metadata – Audio object structures for preservation and restoration.

3.2 Instance Document

A document that conforms to the minimally required set of elements and attributes defined by an XML schema.

3.3 Signal graph

An input through output tracing of all audio signal routing through a series of devices and or software describing the total processing a